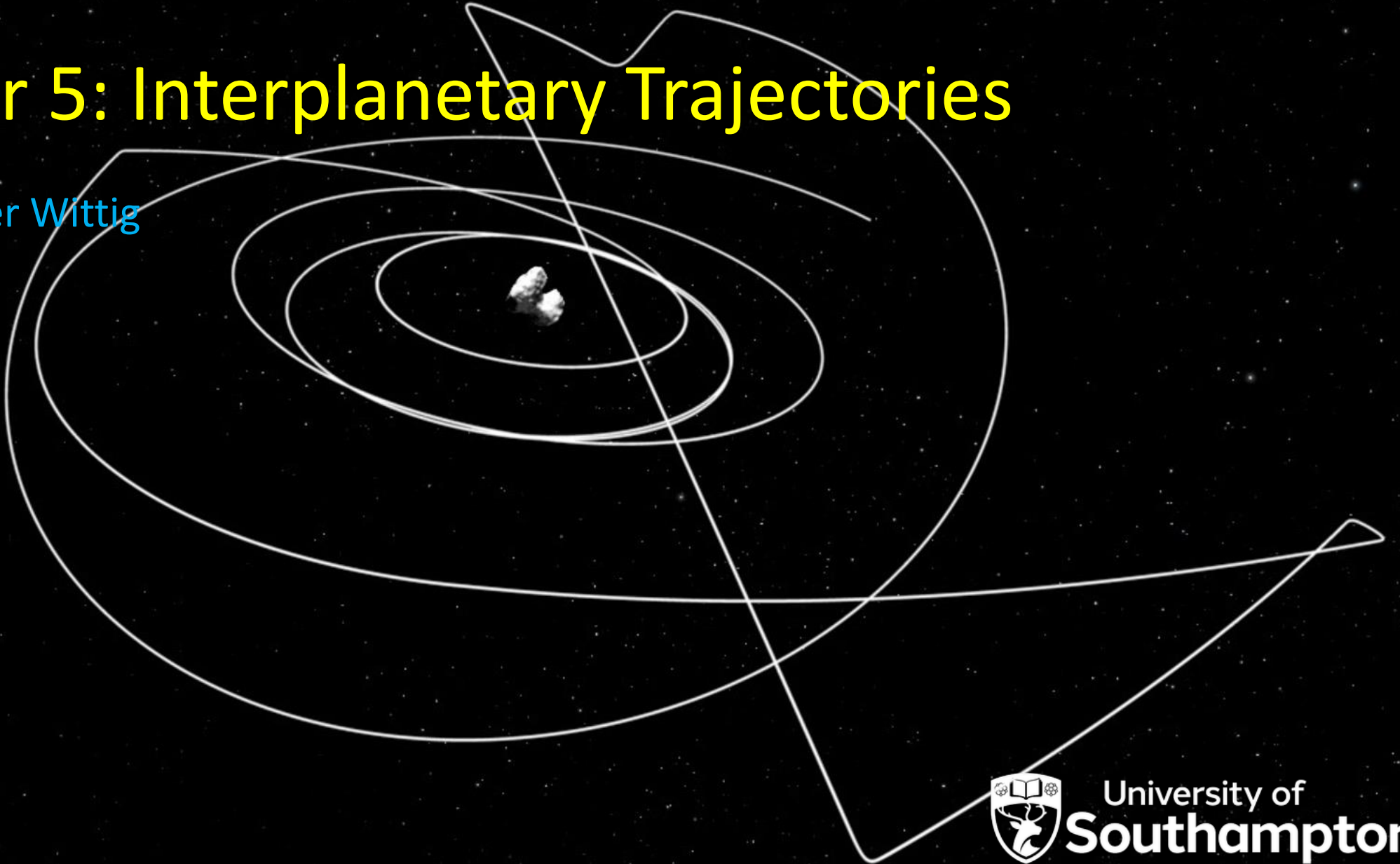


Orbital Mechanics (SESA6076)

Chapter 5: Interplanetary Trajectories

Dr. Alexander Wittig



- Any questions on previous weeks content?

- **Introduction**
- **Chapter 1: In 20.000 km, turn right. Orbital Manoeuvres**
 - Coplanar Manoeuvres
 - Non-Coplanar Transfers
- **Chapter 2: That's mildly disturbing Orbital Perturbations & Propagation**
 - Non-Uniform Gravity Field
 - Third-Body Perturbation
 - Atmospheric Drag
 - Solar Radiation Pressure
 - Orbit Propagation Methods
- **Chapter 3: Is it a plane? Is it a bird? Orbit Observation & Determination**
 - Observation Techniques
 - Preliminary Orbit Determination
- **Chapter 4: Waltzing Matilda Relative Orbital Motion**
 - Relative Orbital Motion
 - Linearization & State Transition Matrix
- **Chapter 5: To boldly go... Interplanetary Trajectories & Optimization**
 - Lambert's Problem
 - Patched Conics
 - Gravity-assisted fly-by trajectories
- **Chapter 6: Torn between two bodies Restricted Three-Body Problem**
 - N-Body Problem
 - Circular Restricted Three-Body Problem
 - Natural Dynamics around Libration Points
- **Chapter 7: You spin me right round Attitude Dynamics**
 - Attitude representation
 - Attitude Dynamics
- **Chapter 8: In the spin cycle Attitude Motion**
 - Torque-Free Motion
 - Attitude Maneuvers
- **Summary and Revision**

- Perform preliminary mission analysis on interplanetary trajectories
- Compute launch windows for interplanetary Hohmann transfers with phasing
- Apply the method of patched conics to trajectory planning
- Design gravity assist fly-by maneuvers

Interplanetary Hohmann Transfers & Phasing

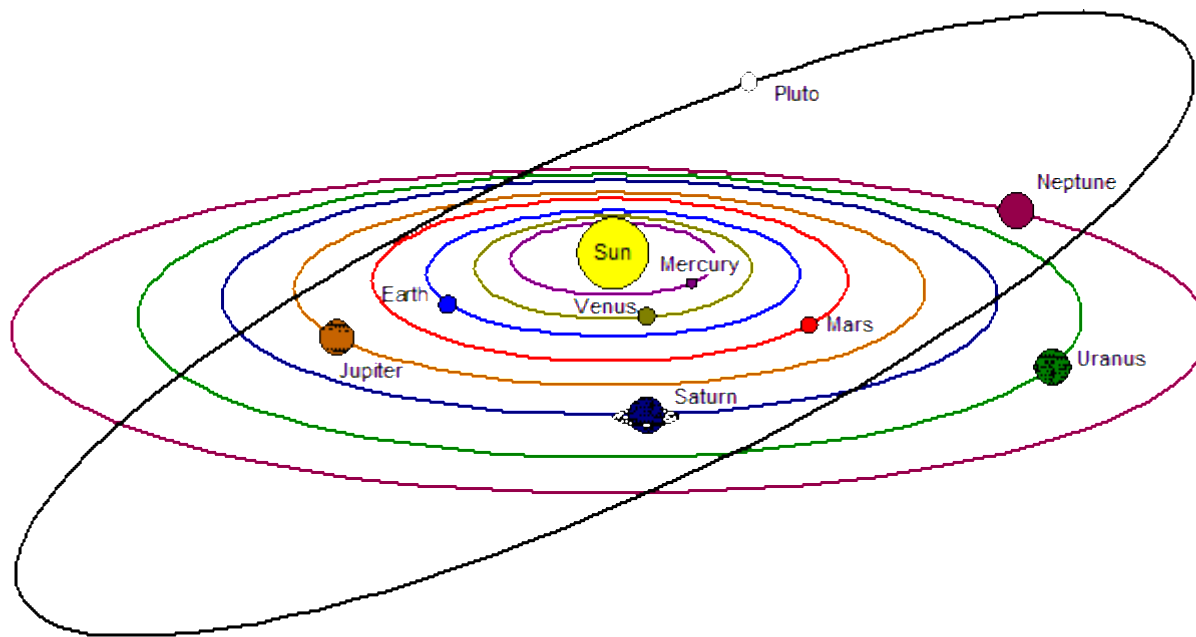


Table A.1 Astronomical Data for the Sun, the Planets and the Moon

Curtis

Object	Radius (km)	Mass (kg)	Sidereal Rotation Period	Inclination of Equator to Orbit Plane	Semimajor Axis of Orbit (km)	Orbit Eccentricity	Inclination of Orbit to the Ecliptic Plane	Orbit Sidereal Period
Sun	696,000	1.989×10^{30}	25.38d	7.25°	—	—	—	—
Mercury	2440	330.2×10^{21}	58.65d	0.01°	57.91×10^6	0.2056	7.00°	87.97d
Venus	6052	4.869×10^{24}	243d*	177.4°	108.2×10^6	0.0067	3.39°	224.7d
Earth	6378	5.974×10^{24}	23.9345h	23.45°	149.6×10^6	0.0167	0.00°	365.256d
(Moon)	1737	73.48×10^{21}	27.32d	6.68°	384.4×10^3	0.0549	5.145°	27.322d
Mars	3396	641.9×10^{21}	24.62h	25.19°	227.9×10^6	0.0935	1.850°	1.881y
Jupiter	71,490	1.899×10^{27}	9.925h	3.13°	778.6×10^6	0.0489	1.304°	11.86y
Saturn	60,270	568.5×10^{24}	10.66h	26.73°	1.433×10^9	0.0565	2.485°	29.46y
Uranus	25,560	86.83×10^{24}	17.24h*	97.77°	2.872×10^9	0.0457	0.772°	84.01y
Neptune	24,760	102.4×10^{24}	16.11h	28.32°	4.495×10^9	0.0113	1.769°	164.8y
(Pluto)	1195	12.5×10^{21}	6.387d*	122.5°	5.870×10^9	0.2444	17.16°	247.7y

*Retrograde

- mostly planar except Pluto & Mercury (17°, 7°)
- mostly circular (same exceptions + Mars)

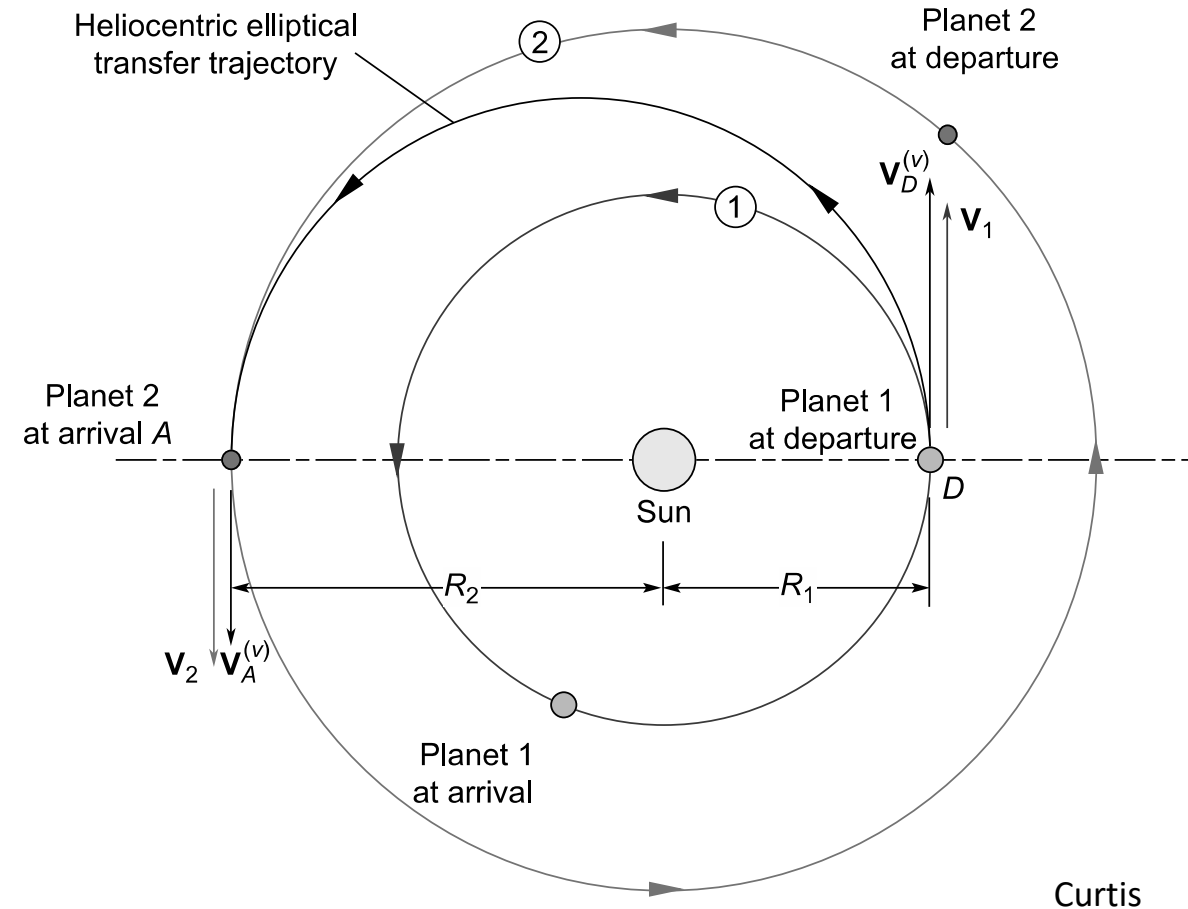
Interplanetary Hohmann

- Most energy efficient transfer:
Classical Hohmann transfer
- Same as before, but now calculated
around Sun
- Assume all orbits to be circular, coplanar

Recall:

$$\Delta v_{Dep} = \sqrt{\frac{\mu_{Sun}}{R_1}} \left(\sqrt{\frac{2R_2}{R_1 + R_2}} - 1 \right)$$

$$\Delta v_{Arr} = \sqrt{\frac{\mu_{Sun}}{R_2}} \left(1 - \sqrt{\frac{2R_1}{R_1 + R_2}} \right)$$



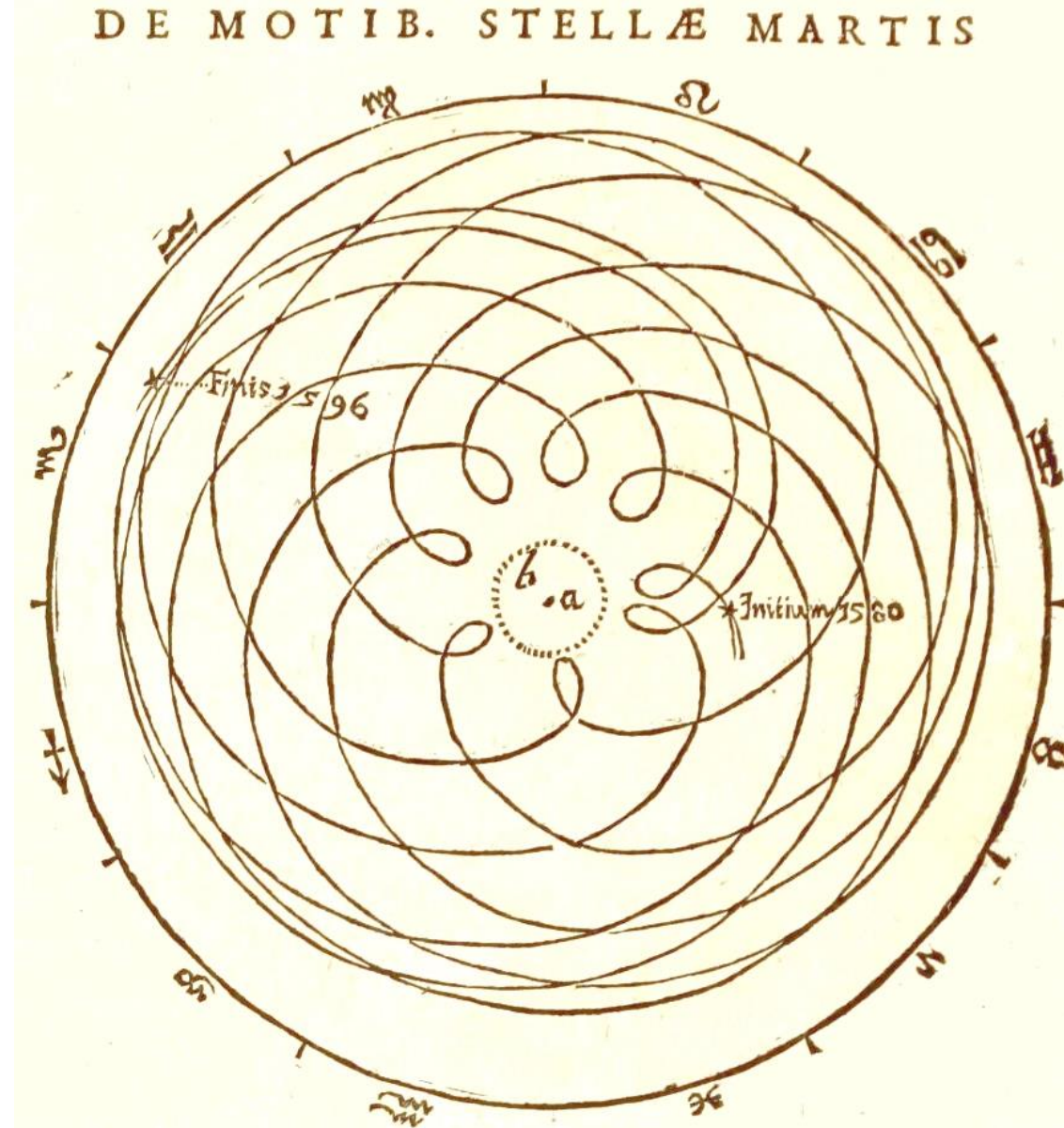
Synodic period

Time after which the same geometric constellation of some celestial bodies repeats.

- May be rotated

Transfer timing:

- What is required angle for outward trip?
- What is required angle for inward trip?
- How long is wait time for round trip?



Synodic period

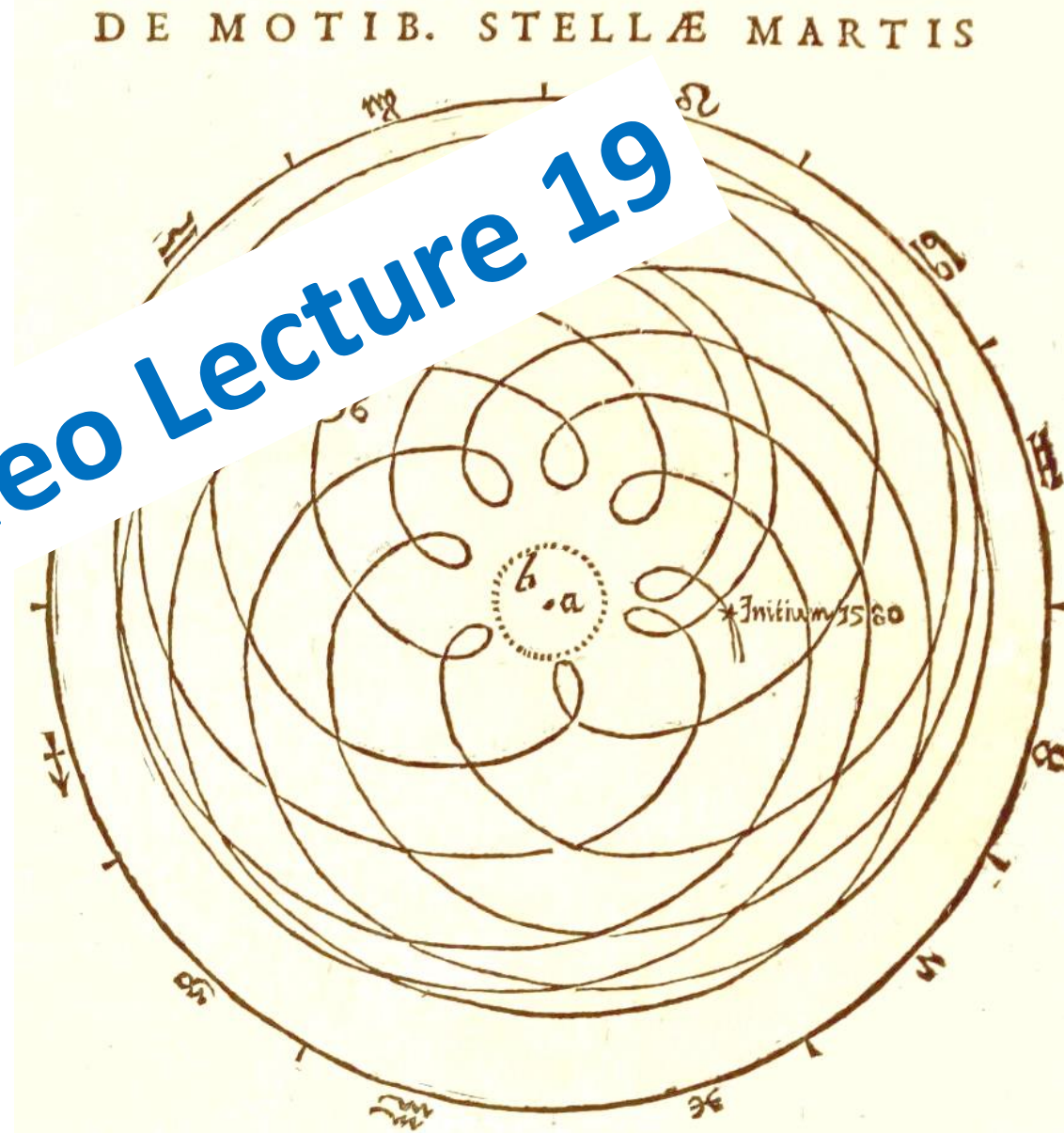
Time after which the same geometric constellation of some celestial bodies repeats.

- May be rotated

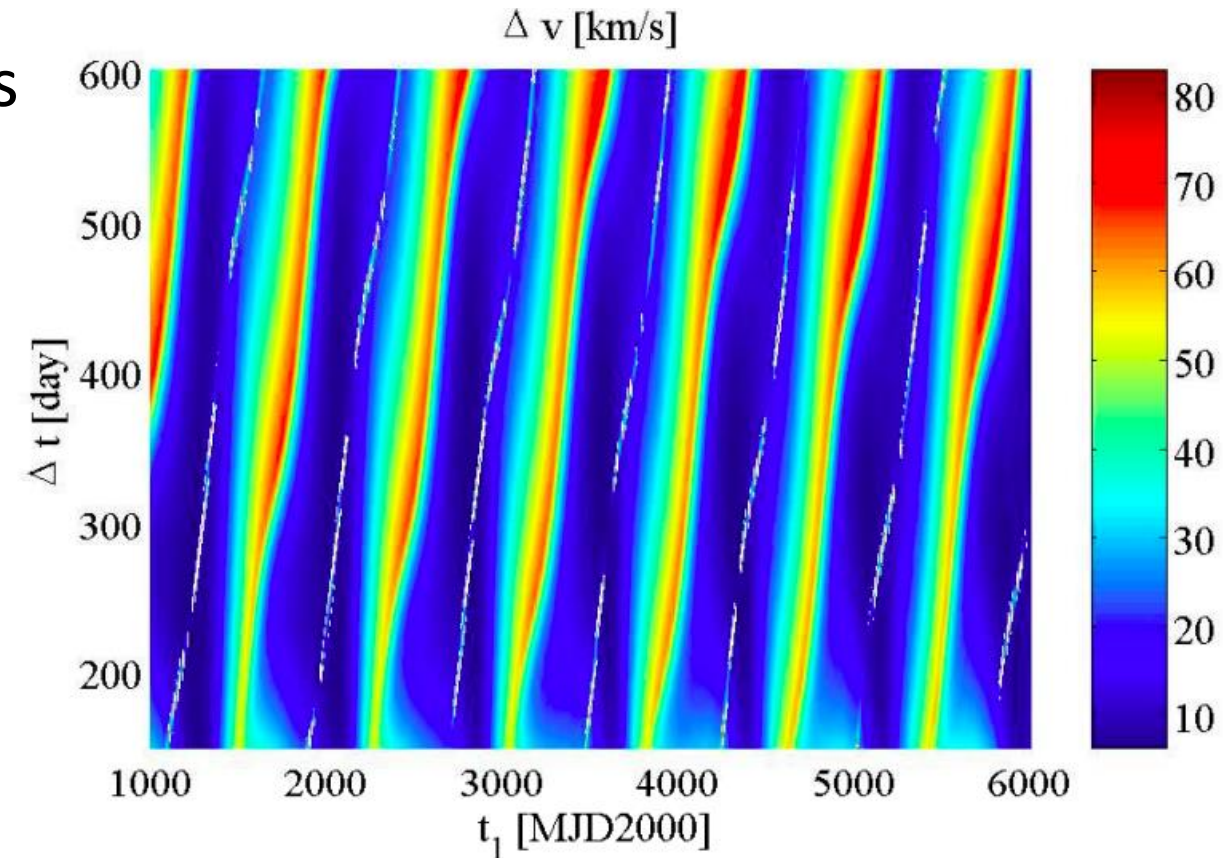
Transfer timing:

- What is required for outward trip?
- What is required for inward trip?
- How long is the time for round trip?

see Chapter 6 Video Lecture 19

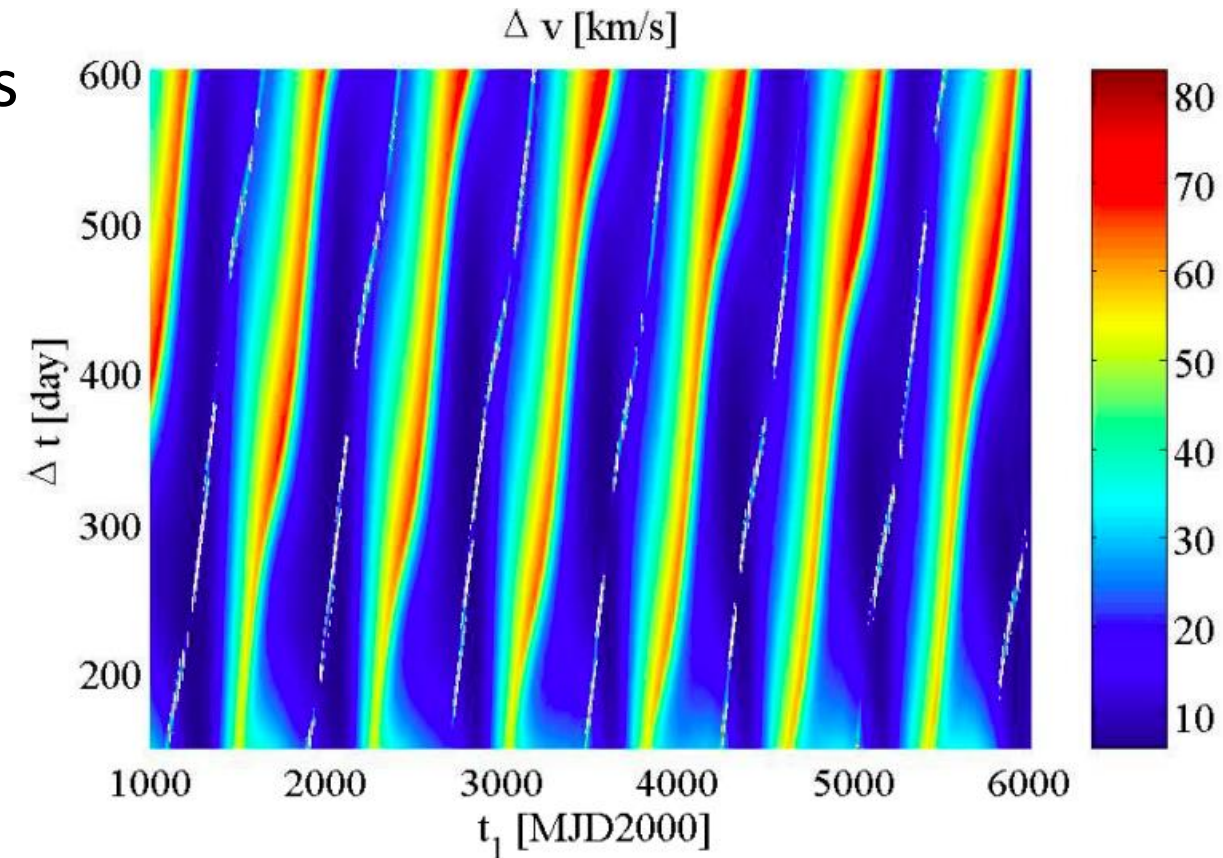


- Extended Hohmann transfer also works
 - with eccentric orbits
 - with orbit inclination

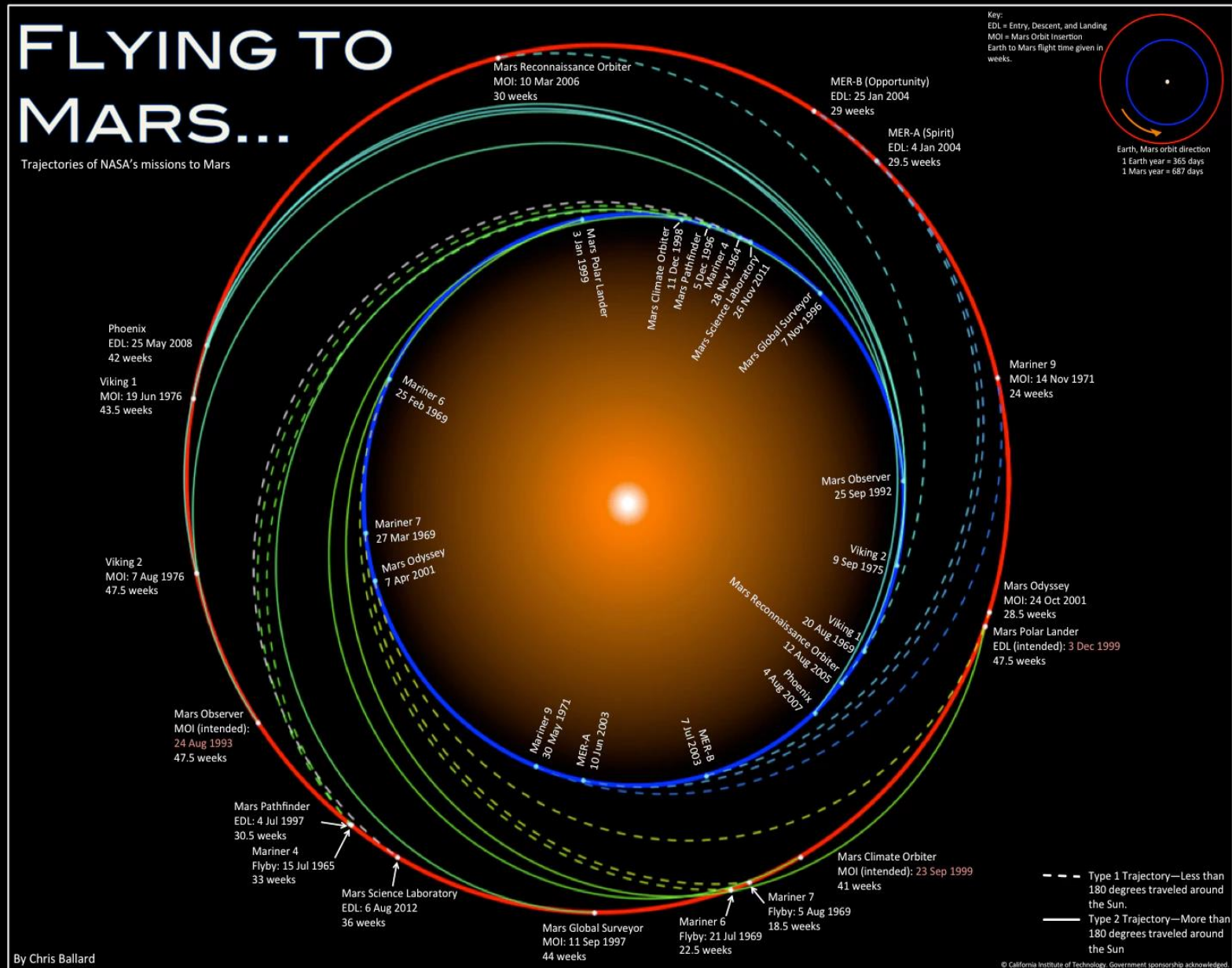


Rigorous global optimization of impulsive planet-to-planet transfers in the patched-conics approximation
R. Armellin, P. Di Lizia, K. Makino and M. Berz

- Extended Hohmann transfer also works
 - with eccentric orbits
 - with orbit inclination
- Problems:
 - Phasing restricts window of opportunity
 - Travel time (radiation, exposure, risk, ...)
- Better: Lambert transfer
 - cost depends on travel time & relative orientation
 - can be prohibitive!



Rigorous global optimization of impulsive planet-to-planet transfers in the patched-conics approximation
R. Armellin, P. Di Lizia, K. Makino and M. Berz





Career:

- MS aerospace engineering (University Southern California, 1965)
- Research engineer (JPL, 1966)
- Head of Mars Exploration Program (NASA, 1994)

Achievements:

- Mission analyst & project manager Mariner 10
- Project engineer on Cassini
- Team lead on Mars Pathfinder & Sojourner Rover projects
- First women to lead a NASA program

Patched Conics

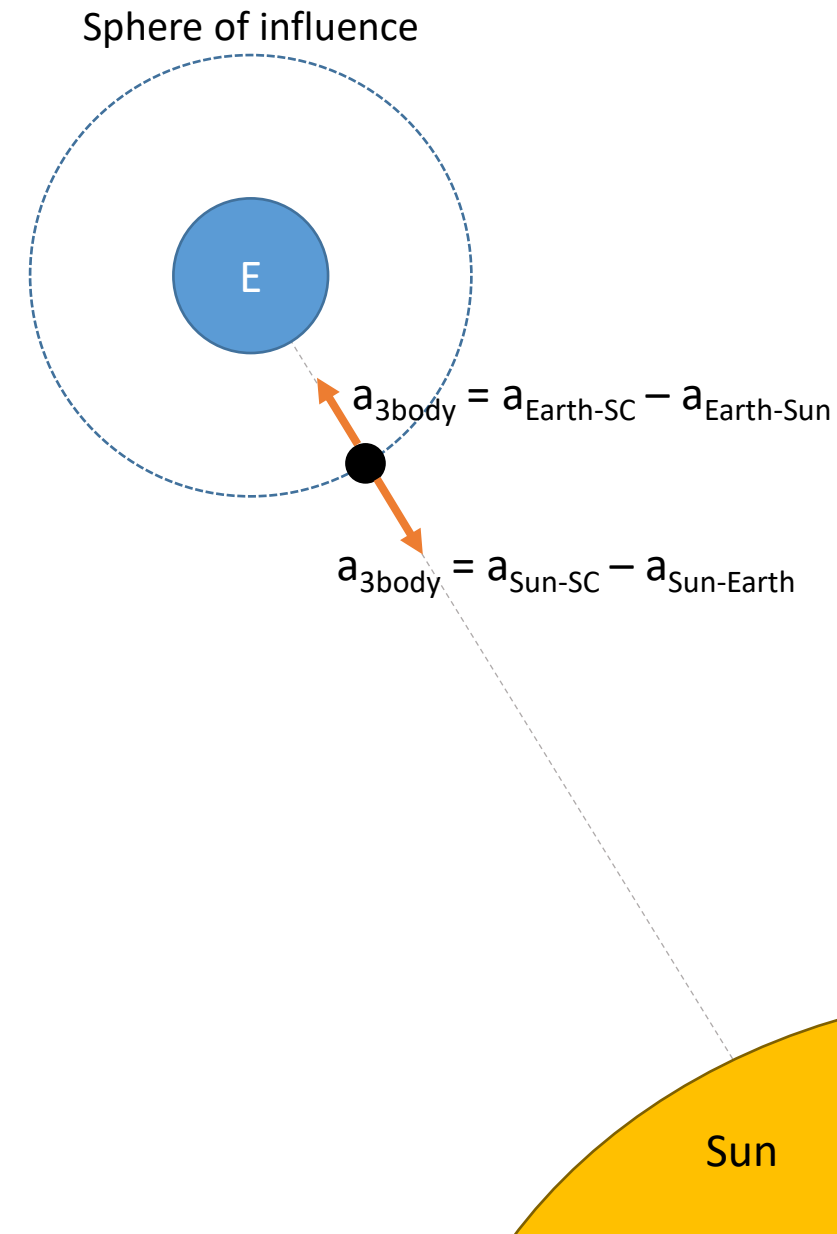
- Sun as third body perturbation?
- Sun as central body?

When is Sun a perturbation, and when main body?

- Fully correct: three body problem (next Chapter)
- Approximation: **Sphere of Influence (Sol)**

Definition:

At intersection of Sol and line between Sun and planet:
perturbation from Sun on Earth-S/C system equals
perturbation from Earth on Sun-S/C system in magnitude.



- From definition & Newton's laws:

$$\frac{r_{SOI}}{R} = \left(\frac{m_p}{m_s} \right)^{\frac{2}{5}}$$

m_p = mass planet m_s = mass Sun
 R = distance Planet-Sun

- Rough estimate: not a hard barrier in space
 - Within: Keplerian + perturbation around body
 - Without: Keplerian + perturbation around Sun

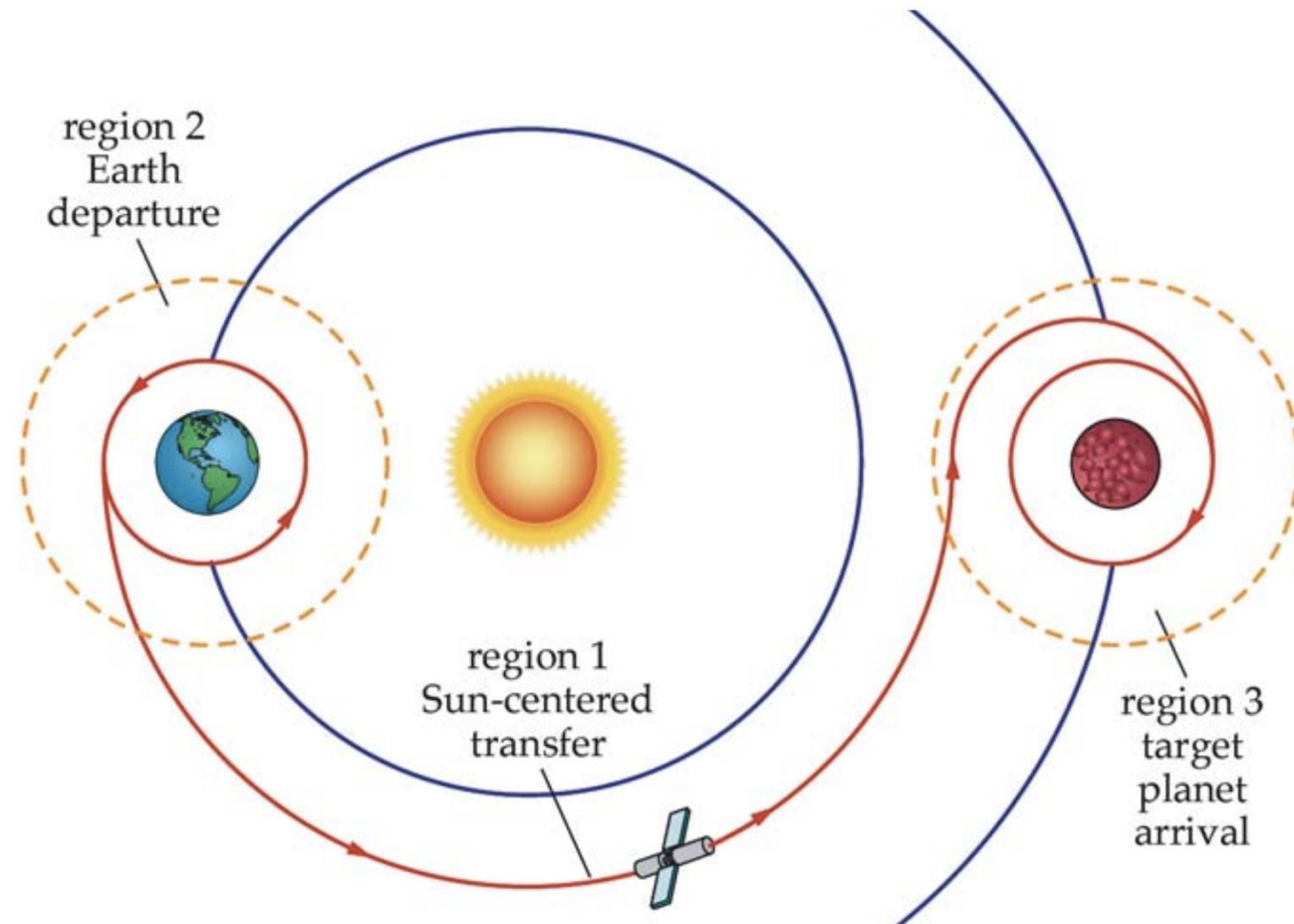
Body	Sol (10 ⁶ km)	Sol (radii)
Mercury	0.112	46
Venus	0.616	102
Earth	0.924	145
Moon	0.0661	38
Mars	0.576	170
Jupiter	48.2	687
Saturn	54.6	1025
Uranus	51.8	2040
Neptune	86.8	3525

Patched Conics Method for preliminary mission analysis

- Patched: trajectory from piecewise Keplerian orbits “patched” together
- Conics: Keplerian orbits around (different) central bodies

Examples:

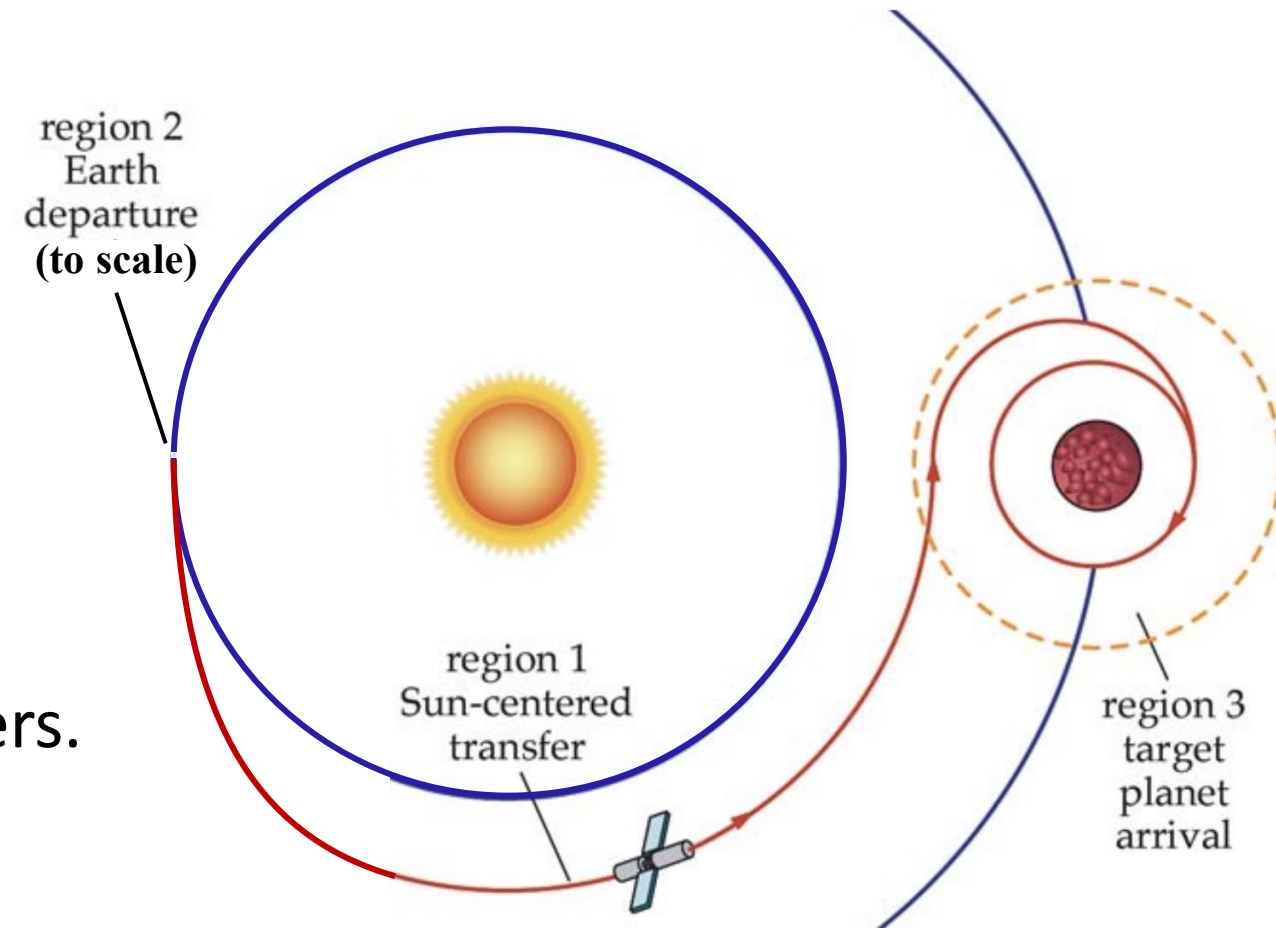
- | | |
|------------------------|---------------|
| 1. Earth-Mars transfer | Sun-centric |
| 2. Earth departure | Earth-centric |
| 3. Mars arrival | Mars-centric |



Patched Conics Method for preliminary mission analysis

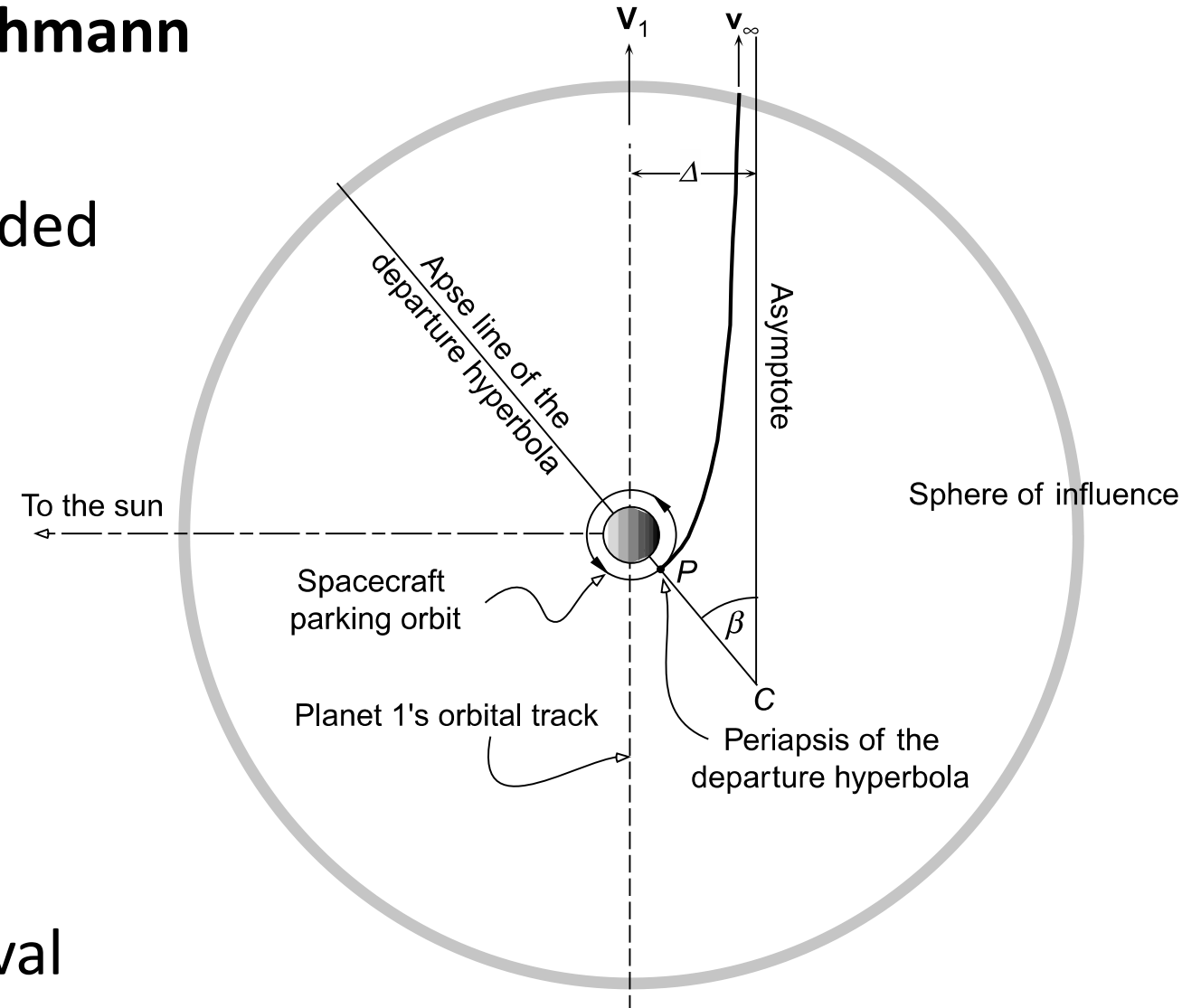
- Leaving Sol $\approx$ Infinity
- $r_{\text{Sol}} \ll r$ for planets
e.g. Earth: $0.9 \ll 150$ [10^6 km]

Departure / Arrival treated as points,
approximately instantaneous maneuvers.



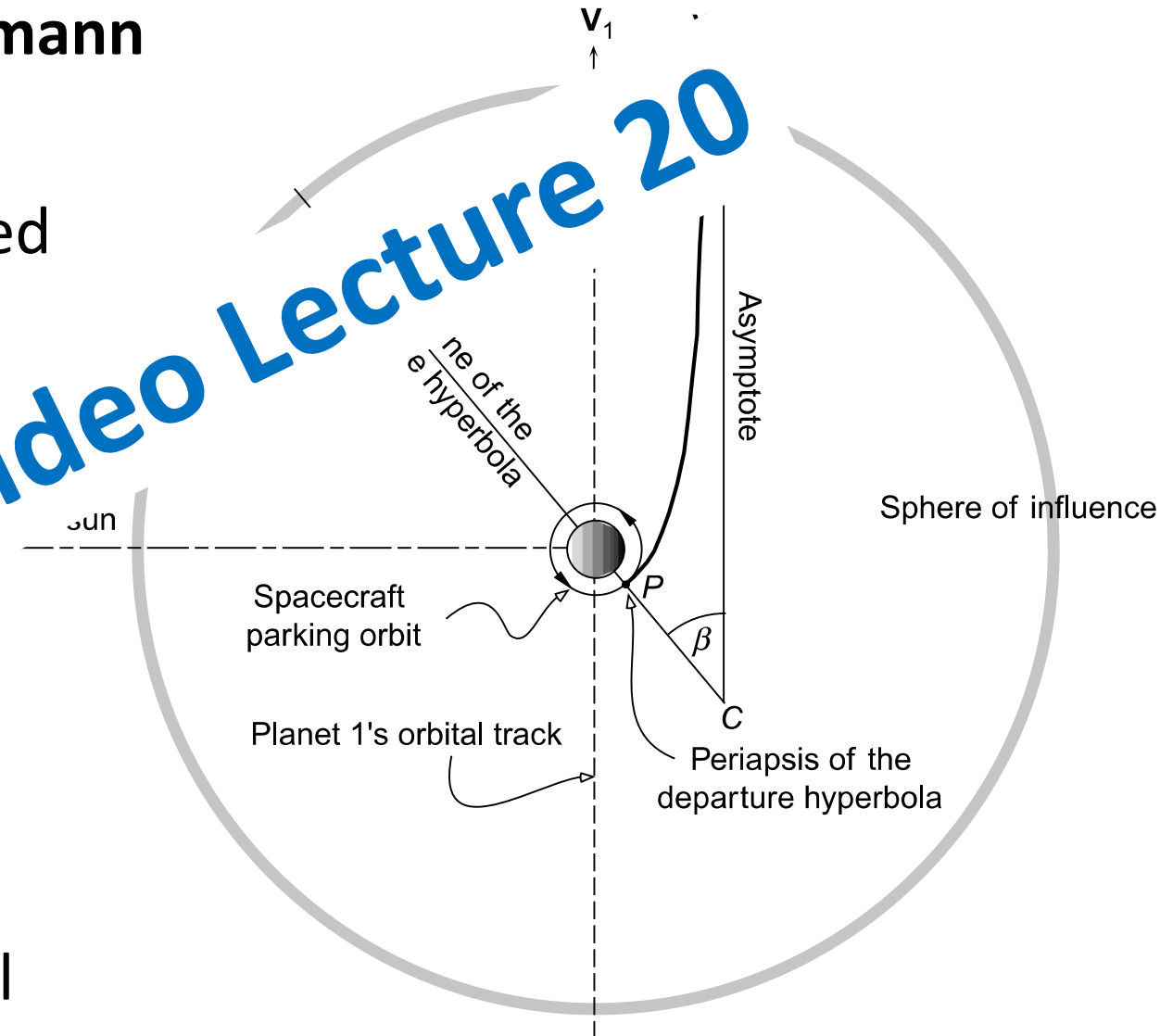
Planet Departure into heliocentric Hohmann

- Leave planet by switching from bounded parking orbit to hyperbolic escape
- To minimize ΔV , align with direction of planet motion around Sun
- Once Sol reached, switch over to Sun
- Similar but backwards for planet arrival



Planet Departure into heliocentric Hohmann

- Leave planet by switching from bounded parking orbit to hyperbolic escape
- To minimize ΔV , align with direction of planet motion around Sun
- Once SoI reached, fly over to Sun
- Similar but backwards for planet arrival



- Once escaped from Earth: transfer relative to Sun
 - Hohmann transfer
 - Lambert arc
- Earth departure is very sensitive to errors
 - 1.1 m/s error in v_p misses Mars by about 153,000 km
 - 670 m error in r_p misses Mars by about 70,000 km

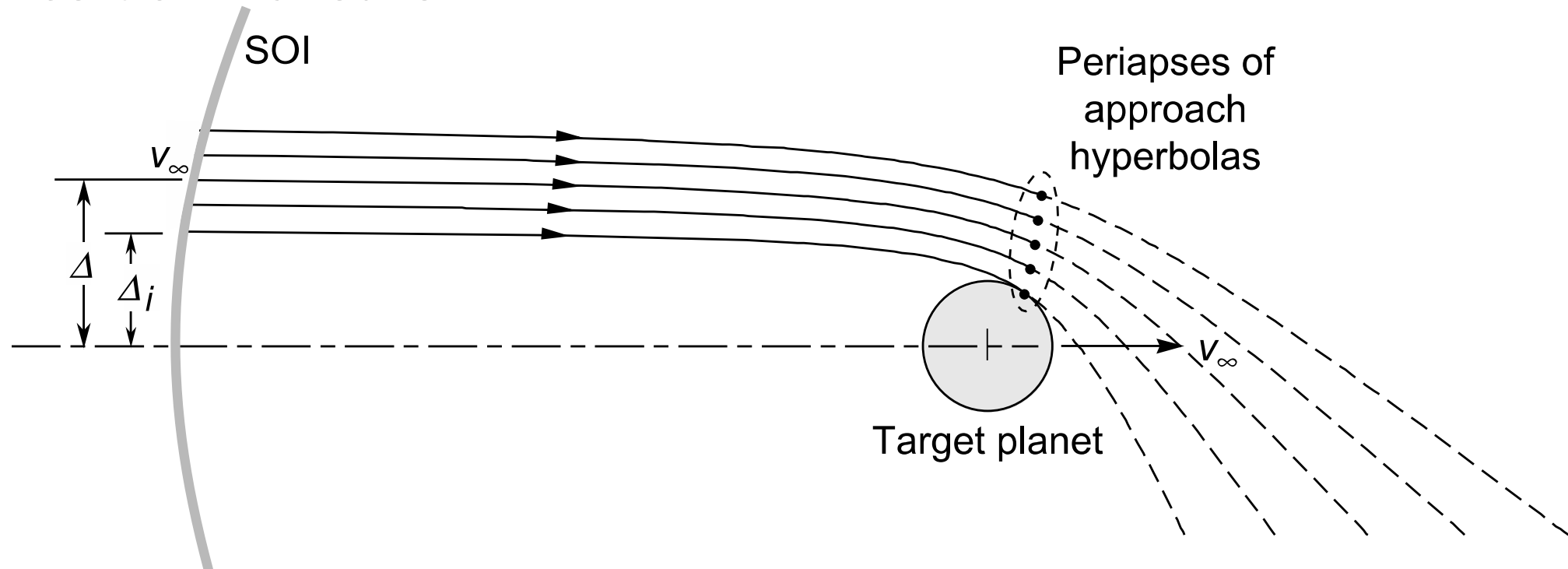
Solution: Mid-course correction maneuver as needed

- Escape and arrival / fly-by are fast (order of days)
- **Sun centric part(s) of trajectory sufficient to estimate mission time**

Scattering problem

- v_∞ and aiming radius Δ define arrival hyperbola
- No maneuver: flyby, leaving planet again (capture not possible in 2BP!)
- Need insertion maneuver

When missing SOI entirely, nothing happens



High precision time-resolved thrust measurement and Two-dimensional plasma plume diagnostics

Project Description

This exciting PhD project will develop a high precision time-resolved thrust balance and optical diagnostics to characterise the spatial distribution of plasma plume generated by a plasma thruster. It will contribute developing a novel plasma thruster, called ALPS (ALI-printed Propulsion System) by:

- **Characterising plasma plumes and their interactions:**
- **Measuring thrust under wide operating conditions**

This project is also part of the UK Space Agency's National Space Innovation Programme (NSIP), offering you the opportunity to contribute to real-world thruster development.

Requirement

- 1st class MEng (or equivalent)
- UK / EU applicants are preferred.
- Highly talented international (non-EU/UK) applicants are accepted
- Having basic engineering design skills

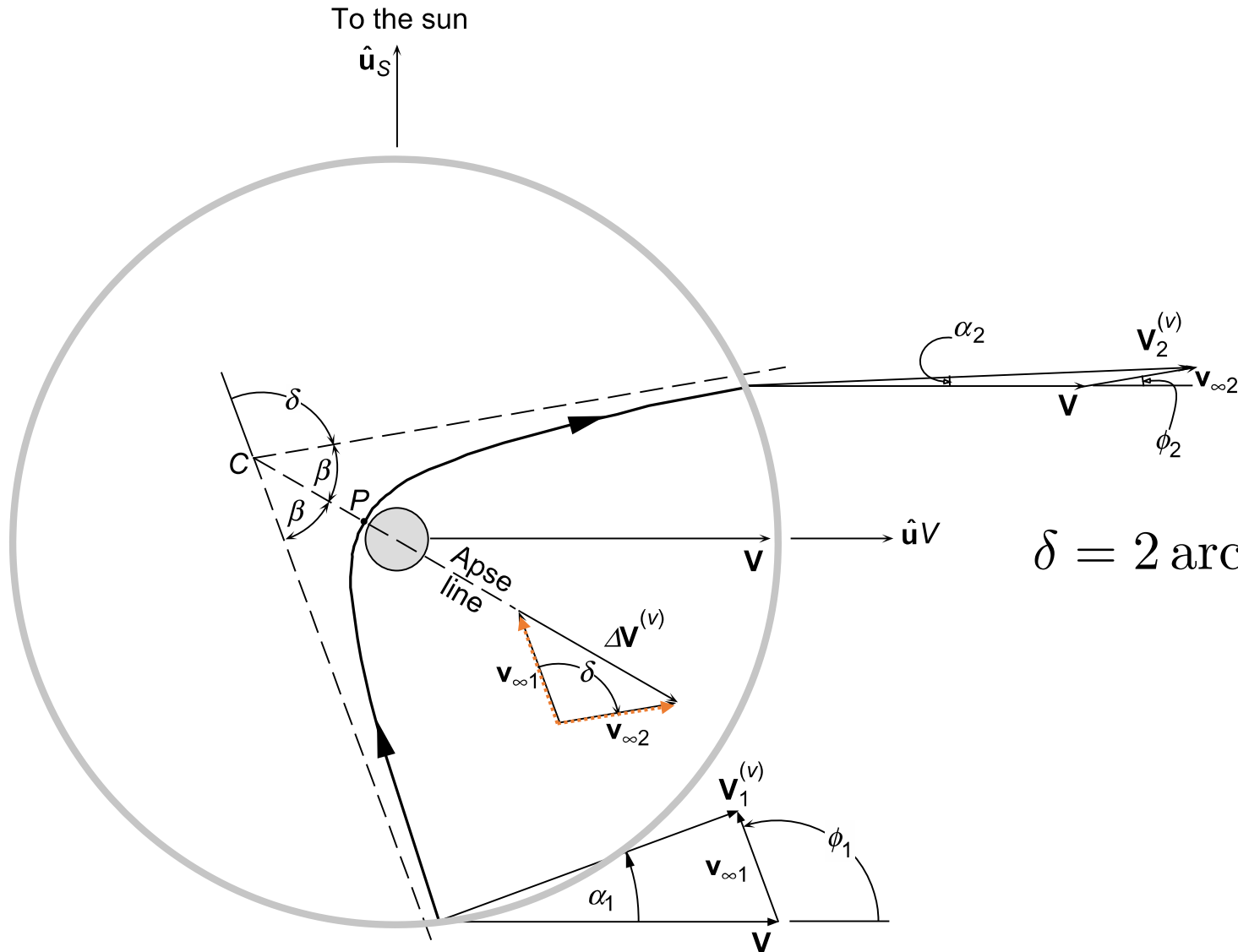
Contact: Dr Minkwan Kim (m.k.kim@soton.ac.uk) with your CV and transcripts

Gravity assist maneuvers



ESA JUICE mission

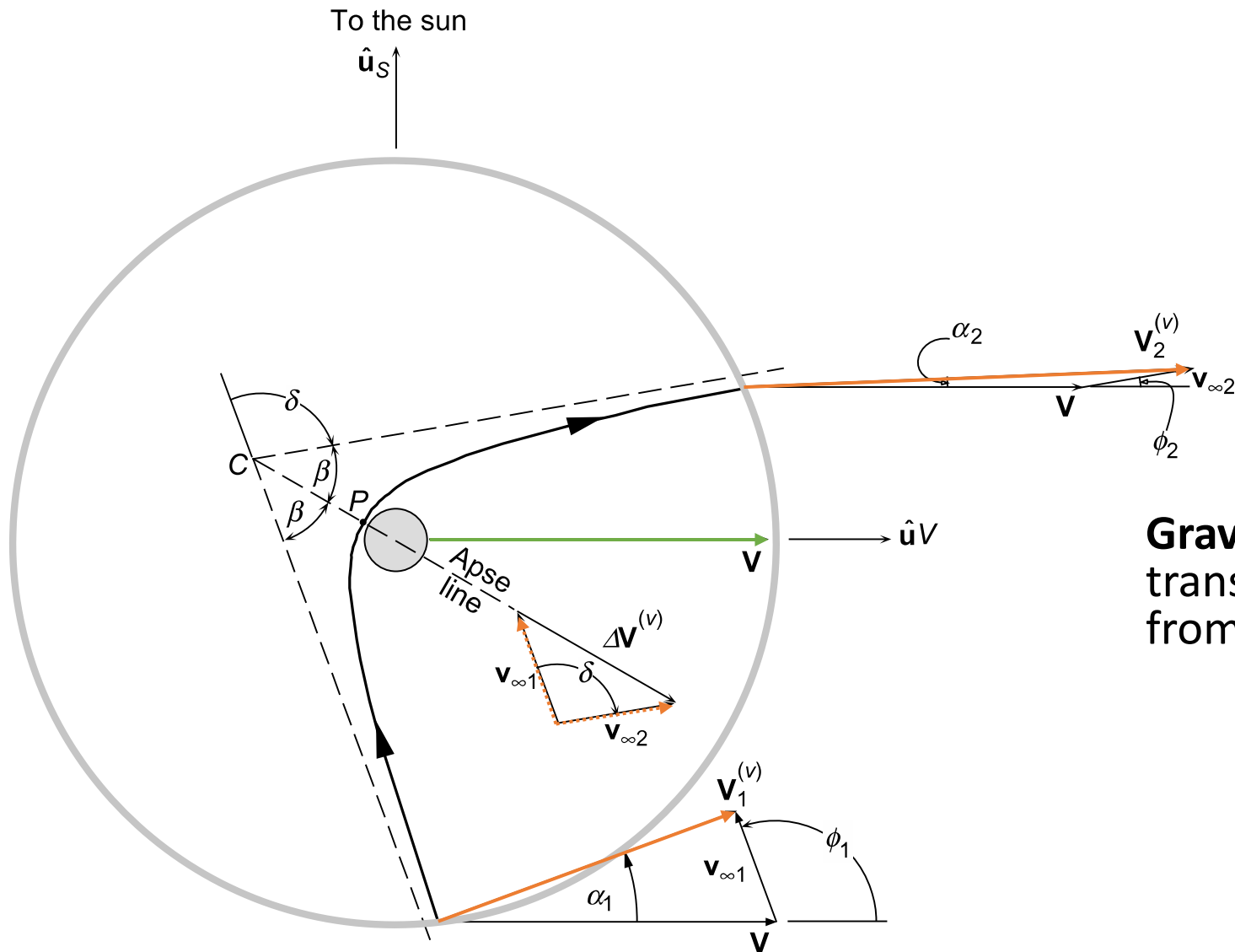
Mechanism of Gravity Assists



- Same hyperbolic fly-by
- Velocity **only rotated** in hyperbolic fly-by
- $\mathbf{v}_{\infty,1} = \mathbf{v}_{\infty,2} = \mathbf{v}_{\infty}$

$$\delta = 2 \arcsin \left(\frac{1}{e} \right) = 2 \arcsin \left(\frac{1}{1 + \frac{r_p v_{\infty}^2}{\mu}} \right)$$

Where does energy gain come from?



- Same hyperbolic fly-by
- Velocity **only rotated** in hyperbolic fly-by
- $V_{\infty,1} = V_{\infty,2} = V_{\infty}$

Gravity assists or **Swing-by** maneuvers transfer energy and angular momentum from a planet to the spacecraft.

- Arbitrary approach angle relative to planet velocity:
compute $\mathbf{v}_{\infty}$ vectorially
- Change of orientation of $\mathbf{v}_{\infty}$ relative to $\mathbf{v}_{\text{planet}}$!



Fly-by is considered an impulsive (instantaneous) maneuver changing only spacecraft velocity at position of planet.

- Usually planar, but can also be used out of plane (plane change)
- Typically optimized numerically, with Lambert transfers between planets
- Limited by safe radius of planet

Outline: given r_p (or Δ) find $\mathbf{v}_{sc,2}$

1. Compute incoming sun-centric velocity vectors $\mathbf{v}_{sc,1}$ and $\mathbf{v}_{planet}$
2. Compute $\mathbf{v}_{\infty,1}$ and deflection angle δ , and rotate $\mathbf{v}_{\infty,1}$ into $\mathbf{v}_{\infty,2}$
3. Add $\mathbf{v}_{planet}$ to obtain $\mathbf{v}_{sc,2}$

Numerical Mission Optimization

Missions made from small building blocks

- So far: various small maneuvers/operations
- Combine building blocks appropriately
 - analytical
 - numerical
- Internals of building blocks: solved separately
 - consider solved / leave for later / use from library
- “Divide and Conquer” strategy
- Inputs/Output:
 - positions & velocities (**coordinates!**)
 - times (**absolute/relative**)
 - Delta-V, other parameters



- (Keplerian) propagation about a body

$$x_f, v_f = \textit{Kepler}(x_i, v_i, \Delta t)$$

- Delta V maneuver

$$v_f = v_i + \Delta v$$

- Fly-by at a body

$$v_f = \textit{Flyby}_{body}(t, v_i, r_p)$$

- Lambert Transfer

$$v_{dep}, v_{arr} = \textit{Lambert}(x_i, x_f, \Delta t)$$

- Ephemeris of celestial objects

$$x_{body}, v_{body} = \textit{Eph}_{body}(t)$$



Idea: Formulate problems as **optimization problem**

- Optimization variables x
- Objective function $f(x)$
 - single objective / multi objective
 - constrained / unconstrained
- Constraint functions g, h
 - equality constraints
 - inequality constraints
 - domain constraints (on x)

$$\begin{aligned} &\text{minimize } f(x) \\ &\text{subject to } g_i(x) \leq 0 \text{ for each } i \in \{1, \dots, m\} \\ &\quad h_j(x) = 0 \text{ for each } j \in \{1, \dots, p\} \\ &\quad x \in X. \end{aligned}$$

Example: calculate $x=15.4/13.9 \rightarrow$ minimize $(13.9*x - 15.4)^2$
(only uses multiplication!)

Idea: Formulate problems as **optimization problem**

- Optimization variables x
- Objective function $f(x)$
 - single objective / multi objective
 - constrained / unconstrained
- Constraint functions g, h
 - equality constraints
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$$\begin{aligned} &\text{minimize } f(x) \\ &\text{subject to } g_i(x) \leq 0 \text{ for each } i \in \{1, \dots, m\} \\ &\quad h_j(x) = 0 \text{ for each } j \in \{1, \dots, p\} \\ &\quad x \in X. \end{aligned}$$

Challenge: formulate mission design problem correctly

- GMAT allows to build simple mission profiles
- More complex missions: own code using libraries
 - ESA PyKep
 - NAIF SPICE
 - ...
- Optimizers (non-linear programming):
 - Matlab (e.g. fmincon)
 - Python / SciPy (e.g. scipy.optimize.minimize)
 - SNOPT
 - ...

Need to define:

- optimization variables $\mathbf{x}$ (and their domain)
- objective function $\mathbf{f}$
- constraints $\mathbf{g}, \mathbf{h}$

Independent of tools: must develop objective function to match mission

Typical objectives in Mission Analysis (MA)

- Minimize
 - **Delta-v** (most common)
 - Fuel
 - Mission Duration
 - Maneuvers
- Maximize
 - Payload
 - Profit
 - Science

Typical constraints

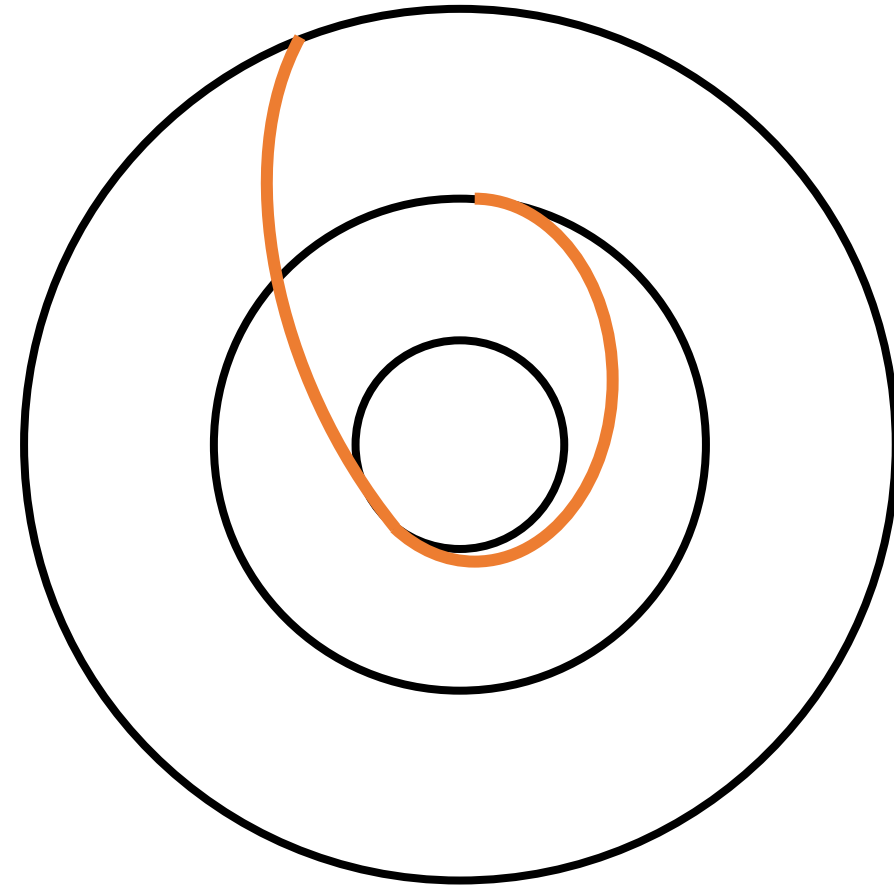
- Inequality
 - mission duration
 - geometrical constraints (forbidden regions)
 - maximum thrust
- Equality
 - prescribed positions
 - patch points
 - rendezvous times

Mission profile:

- start at Earth at given t_0
 - up to 3 km/s v_∞ in any direction free from launcher
- one fly-by at Venus on the way
- arrival at Saturn, matching its velocity

Need to define:

- optimization variables $\mathbf{x}$ (and their domain)
- objective function $\mathbf{f}$
- constraints $\mathbf{g}, \mathbf{h}$



Example: Transfer with Fly-By

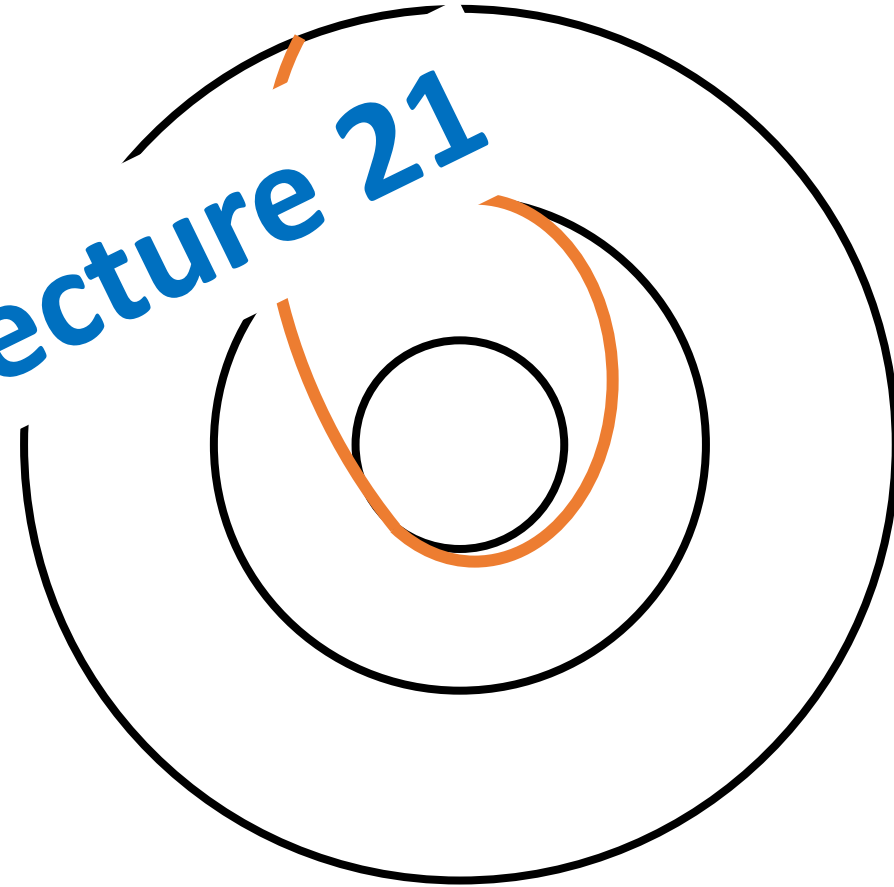
Mission profile:

- start at Earth at given t_0
 - up to 3 km/s v_∞ in any direction free from launcher
- one fly-by at Venus on the way
- arrival at Saturn, matching its velocity

Need to define:

- optimization variables $\mathbf{x}$ (and their domain)
- objective function f
- constraints $\mathbf{g}, \mathbf{h}$

see Chapter 6 Video Lecture 21



- Hohmann transfer combined with phasing allows right timing:
Launch window
- Sphere of influence is a measure for where a planet acts as primary
- Patched Conics combines successive Keplerian orbits around different bodies
- Can be used for direct transfers or multiple intermediary gravity assist fly-by maneuvers
- Missions made from building blocks with numerically optimized parameters

Further reading

- Curtis, Chapter 8

Also:

- Vallado, Chapters 11 & 12

Orbital Mechanics (SESA6076)

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